COMP 361
Assignment 1 - Artifacts
Due: October 2, 2026, on myCourses at 23:55

TA Office Hours during the week of Sept 30 to October 6, 2026

Introduction

Each assignment in this course has two sections: Individual Exercises and Team Exercises. Complete the individual exercises independently. Complete the team exercises collaboratively with your team.

This assignment is marked out of 20 points.

Individual Exercises

Do this section on your own.

When eliciting the requirements from a client, the engineer meets with the client to discover what they want. In this meeting(s), the engineer creates Artifacts to ground the conversation and make clear the client's needs. In this assignment, I am asking you to create a few Artifacts based on a client's Story.

Use UMLet to create these artifacts.

EXERCISE #1 – Requirements Gathering [16 points]

Submit the following Artifacts for this assignment:

- UML Use-case diagram for the Story (not the Template)
- User interface mockup (or wireframe) for the punch-in/out program
- Jobs Completed report mockup, and
- Write reflection statements about: Impact, Ethics, and Performance. How will this program impact society or the industry? Are there any positive or negative ethical implications? What can we say about performance requirements and hindrances to the software's performance?

For the following Story:

> The manufacturing company, ABC Industries, produces parts to order for its customers. Each to-order part is called a Job. Customers of ABC Industries call the ABC Sales Rep to order the part they want to build. The Sales Rep enters the order into the company database as a Job. Each Job has an ID number (Job number), a title, a text description, a link to the assembly instructions, and a list of parts. The people who build the parts (handle the Job) are called the Floor Staff. When an individual Floor Staff employee has nothing to do, they go to the Punch-in/out computer to get a Job. They see a list of available jobs (i.e., jobs no

one is working on), also known as the Jobs List. From that list, they select a job and enter their employee number to assign the work to themselves. At this point, the system checks whether the employee is already punched into another job. If so, the system punches them out from the previous job and punches them in for the new job. Another table in the database records how much time each Floor Staff employee has worked on a job. If the employee has worked on the job multiple times and/or multiple days, the system keeps a record for each work event. This event contains the date, start and end time, job number, and employee number. To punch out means the computer finds the current open record in the database and updates the end time. To punch in means appending a new record to the database with the date, job number, employee number, and start time (the end time is left blank). In addition to punching in/out of jobs from the user interface, the Floor Staff employee can also enter a date range to view all the Jobs they worked on (the Jobs Completed Report). The report is divided into days. For each day, it lists the jobs the employee worked on. At the end of the day, it displays the total hours and minutes worked. At the end of the report, it displays the total hours and minutes worked over the entire period.

To summarize, the user of this program (the Punch-in/out program) can select four actions from the computer: punch-in, punch-out, report, and exit program. The punch-in operation will automatically punch out the employee if they are already punched in on another job. The punch-out operation only punches the employee out of a job. The employee should not exit the program, so it should prompt, “Are you sure?” If the user forgot to punch out at the end of the day, the program automatically punches everyone who has not already punched out at 5:30 PM.

Your job is to develop the most well-thought-out, optimal, and clean solution. Note that it is possible the story provides more information than is necessary for a given artifact. Be sure to include only the information that is appropriate for the given artifact (never too much, never too little).

What To Submit

Submit the following to the myCourses Assignment tool, placing your work in the Assignment 1 box:

- PDF of a UML Use-case diagram for the entire Story,
- PDF of a User interface mockup or wireframe for the punch-in/out program **only**,
- PDF of a Jobs Completed report mockup,
- PDF for the Impact Statement reflections,
- PDF of the Ethics Statement reflection, and
- PDF of the Performance Statement reflection.

Format your submissions so that they are easy to read.

Rubric

The terms used here are defined at the end of this assignment.

- UML Use-case diagram
  - 0-1 = Syntax
  - 0-1 = Professionalism & Quality
  - 0-1 = Appropriateness, Gracefulness, Correctness & Completeness
  - 0-1 = BS
- User interface mockup or wireframe
  - 0-1 = Syntax
  - 0-1 = Professionalism & Quality
  - 0-1 = Appropriateness, Gracefulness, Correctness & Completeness
  - 0-1 = BS
- Jobs Completed Report mockup (not wireframe)
  - 0-1 = Syntax
  - 0-1 = Professionalism & Quality
  - 0-1 = Appropriateness, Gracefulness, Correctness & Completeness
  - 0-1 = BS
- The three statements
  - 0-1 = Syntax
  - 0-1 = Professionalism & Quality
  - 0-1 = Appropriateness, Gracefulness, Correctness & Completeness
  - 0-1 = BS

Team Exercises

Do this section with your team members.

EXERCISE #2 – Team Formation and Project Selection [4 points]

For this exercise, you will need to do the following:

1. Assemble your team. Your team must have 7 members.
2. Pick a team captain. This person will communicate with the professor and TA on behalf of the team. Individual team members can also speak with the professor and TA if needed or if personal issues arise.
3. Select the project your team will work on. See the project description. Discuss and reach a consensus.
4. Fill out the Microsoft Form to inform the professor of your decision. ONLY YOUR TEAM CAPTAIN FILLS OUT THE FORM. You do not need to submit anything in the Assignment #1 box in myCourses. You only need to fill out the form. Here is the link: _____ You can fill out the form more than once. The form has instructions about multiple/corrected submissions.
5. The professor will update the Prof & TA Coordinates sheet, the Your TA tab, with the information from the form. Please check this sheet to confirm that the information is correct. Speak to the professor to make corrections.

What To Submit

- Do not submit anything to myCourses. You only need to fill out the form.

Rubric

- If the form was never filled out, then 0 points.
- If the form is partially filled out, then 1-3 points (awarded proportionally).
- If the form is completely filled out, then 4 points.

Definitions:

- Syntax (syntax correctness)
    - Drawings and writing follow the standards presented in class for the drawing and written work. For example, use-case diagrams strictly follow the drawing style presented in class and not the internet.
- Professional (in presentation)
    - Drawings and writings are not only correct, but they are also presented professionally. For example, no drawings are created by hand. All writing is free from grammatical and spelling errors.
- Quality (of the solution)
    - This relates to the student attempting to find a solution that incorporates all of these elements: execution optimality, a simple algorithm, good use of design principles (like design patterns), and a fast running time. In terms of drawing or writing, the solution image fits within a page or less, and the writing is neither overly wordy nor curt.
- Appropriate
    - Appropriateness refers to the fact that the elements that form the solution would be used in the real-life implementation.
    - The student correctly identified the algorithms and design patterns needed for every case.
- Correct
    - Is the student's solution correct?
- Complete (completeness)
    - Did the student fully answer the question, or did they leave elements out of their solution?
    - A solution may have edge cases or additional features that also need description.
- Graceful (graceful execution)
    - The student's solution handles errors and run-time issues so that the program would not crash or hang, alternatively prompting the user.
- BS
    - At times, students will use a lot of language to answer a question, hoping that their uncertainty about the correct response will be hidden within the volume of technical language as they dance around the solution.
    - Other forms of BS include providing multiple solutions to a question in the hopes of receiving partial grades.